CompTIA N+
Network Security, Maintenance & Troubleshooting Procedures

SECTION A – CONCEPT APPLICATION

S1. CPU, RAM and Storage Selection

Question : Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time. What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

Solution : When selecting the hardware, I would first consider the type of work the employees will do. Since programming, web browsing and video conferencing can be used at the same time, I would choose a CPU with enough processing power and multiple cores/threads so the system can handle multitasking.

For RAM, I would check the motherboard's supported RAM type, speed and maximum capacity. I would choose enough RAM so the programs can run together without the system depending too much on virtual memory.

For storage, I would prefer an SSD or NVMe SSD because it provides faster boot and application loading compared with an HDD. I would check whether the motherboard has SATA ports or an M.2 slot for the selected storage.

Finally, I would check CPU socket, chipset and BIOS support, RAM compatibility, storage interface, case size and PSU requirements. Checking the manufacturer's specifications before buying the parts would help make sure the components work together correctly.

S2. Antivirus and System Slowdown

Question : After installing antivirus software on all startup office computers, employees notice that their systems have become slower. Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Solution : Antivirus software runs background services to monitor files, processes and network activity. It uses CPU resources when it scans files and checks running programs.

It also uses RAM for its services, databases and scanning processes. If the computer already has limited RAM, the extra usage can make other applications feel slower.

Antivirus software also performs disk reads and writes when it scans files or updates its virus definitions. If a full scan is running, storage activity can become high and other applications may have to wait for disk access.

Therefore, the slowdown can happen because CPU, RAM and storage resources are being used by the antivirus at the same time. I would check the antivirus scan schedule and system resource usage before deciding whether the antivirus itself is the main problem.

S3. USB Keyboard and Printer Troubleshooting

Question : A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office. Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

Answer : First, I would check whether the USB devices are physically connected correctly and test another USB port. I would also check whether the same keyboard or printer works on another computer.

If the hardware works elsewhere, I would check Device Manager on the affected computer for warning symbols or unknown devices. I would identify the keyboard or printer and check the installed driver.

I would download and install the correct driver from the device manufacturer's website if the current driver is incompatible or corrupted. I would restart the computer and test the device again.

If the problem continues, I would check Windows Update and confirm that the operating system is fully updated. I would also remove the device from Device Manager and reconnect it so Windows can detect it again. Finally, I would check for operating system errors or USB service problems and test with another user account if required.

S4. UPS and Surge Protector

Question : Nexus Innovations experiences frequent power cuts and voltage fluctuations. Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

Answer : A UPS, or Uninterruptible Power Supply, provides temporary battery power when the main electricity fails. This gives the user time to save work and shut down the computer safely instead of losing unsaved data.

A UPS can also help provide more stable power to the connected equipment. This is useful when the office has frequent power cuts or voltage problems.

A surge protector helps protect hardware from sudden increases in voltage. It reduces the chance that a voltage spike will damage components such as the PSU, motherboard and storage devices.

I would use a UPS for important workstations and servers and connect the equipment through suitable surge protection. Regular backups are also important because power protection alone cannot prevent every type of data loss.

S5. DNS and DHCP

Question : New employees at Nexus Innovations can reach internal systems by IP address but not by name, and their devices sometimes fail to obtain a valid IP address. Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

Answer : DHCP automatically provides network settings such as an IP address, subnet mask, default gateway and DNS server address to a client. This allows a new computer to join the network without manually configuring every setting.

DNS translates names into IP addresses. For example, when a user enters the name of an internal application server, DNS should return the server's IP address.

The symptom of being able to reach the application server by IP address but not by name points towards a DNS problem. However, the devices sometimes failing to obtain a valid IP address points towards a DHCP problem as well.

Therefore, I would check both services. I would first verify that DHCP is giving clients valid IP addresses and the correct DNS server address. Then I would test DNS using tools such as nslookup and check the internal DNS records. The name-resolution failure is most directly related to DNS, while the invalid-IP symptom suggests a separate DHCP configuration or availability issue.

S6. Firewall Blocking Internal Application Server

Question : After the IT team installs a new firewall, employees can access the internet but are unable to reach the critical internal application server. Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

Answer : First, I would confirm that the internal application server is running and reachable from the same internal network. I would test the server's IP address and the required service port.

Next, I would check the firewall rules, including source IP, destination IP, protocol and destination port. I would look for a deny rule that matches the employee network and the application server traffic.

I would also check firewall logs because they can show whether the connection is being blocked. A packet capture can be used if the cause is still unclear.

If the firewall is blocking legitimate traffic, I would create or modify a specific allow rule for the required source network, destination server and service port. I would place the rule in the correct order and avoid creating a broad allow rule when a more specific rule is possible. After making the change, I would test the application again and confirm the firewall logs show the expected traffic.

SECTION B – PRACTICAL APPLICATION

Task 1: Workstation Assembly

Answer : Since I did not have access to the physical PC Tool Kit, I would follow the following assembly process and document each step with screenshots if the hardware were available.

1. I would prepare a clean workspace and take proper ESD precautions.
2. I would install the CPU into the correct motherboard socket and check that it is seated correctly.
3. I would apply thermal paste if required and install the CPU cooler. I would connect the cooler fan cable to the CPU_FAN header.
4. I would install the RAM modules into the correct DIMM slots according to the motherboard manual.
5. I would install the M.2 NVMe SSD or SATA storage device, depending on the workstation configuration.
6. I would install the motherboard into the case and secure it using the correct screws and standoffs.
7. I would install the PSU and connect the 24-pin ATX motherboard power and 8-pin CPU power connectors.
8. If a dedicated GPU is required, I would install it in the PCIe x16 slot and connect its power cable.
9. I would connect SATA data/power cables where required.
10. I would connect the front-panel power switch, reset switch and LED connectors, along with front USB and audio headers.
11. I would check that no loose screws or cables are touching the fans.
12. I would perform a final visual inspection before connecting external power.
13. I would power on the workstation and check whether it completes POST.
14. I would enter BIOS/UEFI and confirm that the CPU, RAM and storage are detected correctly.

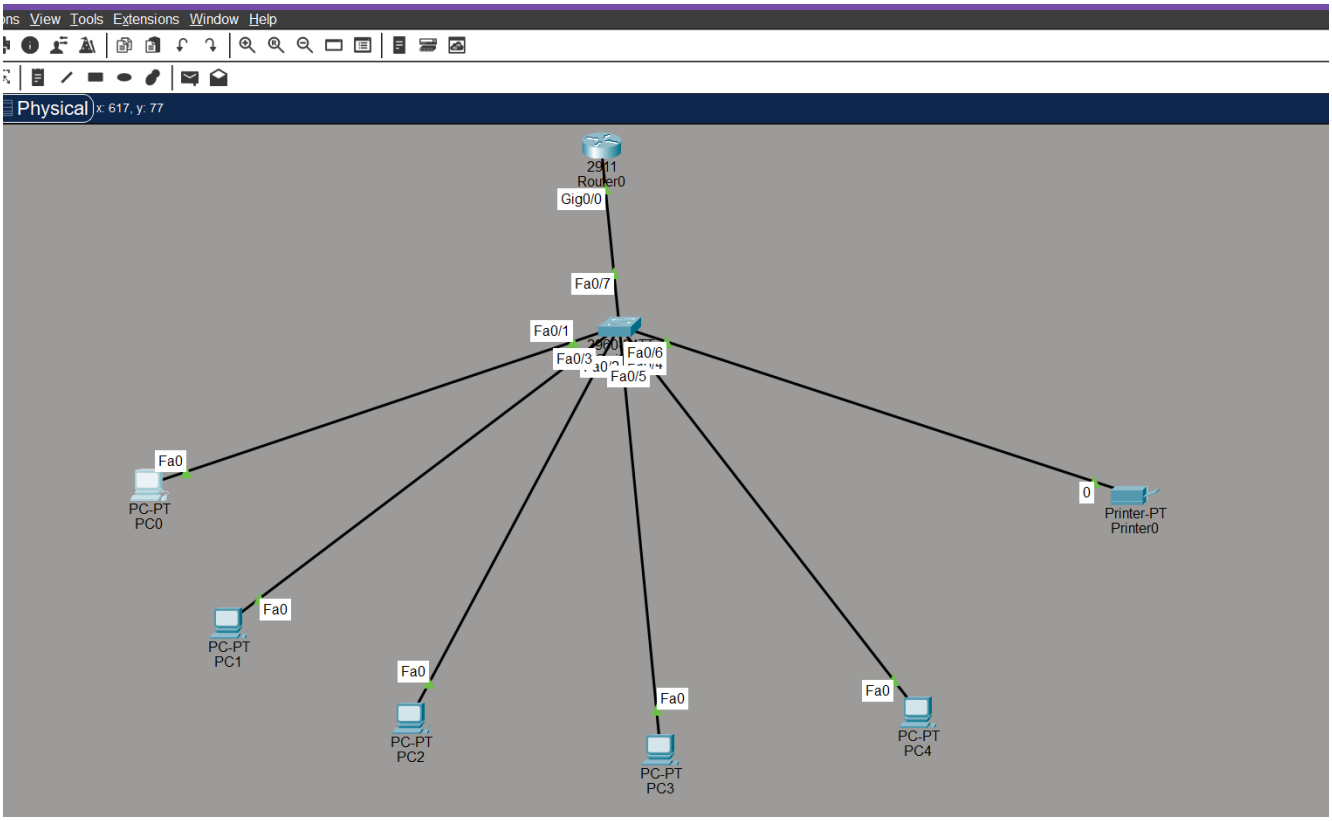
Task 2: RAM Mismatch Troubleshooting

Answer : I would troubleshoot the RAM mismatch through BIOS/UEFI and physical checks as follows.

1. I would restart the workstation and read the BIOS/UEFI POST message or motherboard diagnostic indicator.
2. I would enter BIOS/UEFI and check the detected RAM capacity, speed and configuration.
3. I would compare the installed RAM specifications with the motherboard's supported RAM type, capacity and speed.
4. I would check whether different RAM modules have different speeds, timings or capacities.
5. I would shut down the computer, disconnect power and take ESD precautions.
6. I would reseat the RAM modules and make sure the retaining clips lock into place.
7. If more than one module is installed, I would test one module at a time to identify a faulty or incompatible module.
8. If the modules are compatible but running at an unsupported setting, I would use the normal supported BIOS/UEFI memory settings instead of forcing an incorrect speed or timing.
9. I would save the configuration, reboot and confirm that the system boots normally.
10. Finally, I would check BIOS/UEFI again to confirm the correct RAM capacity is detected.

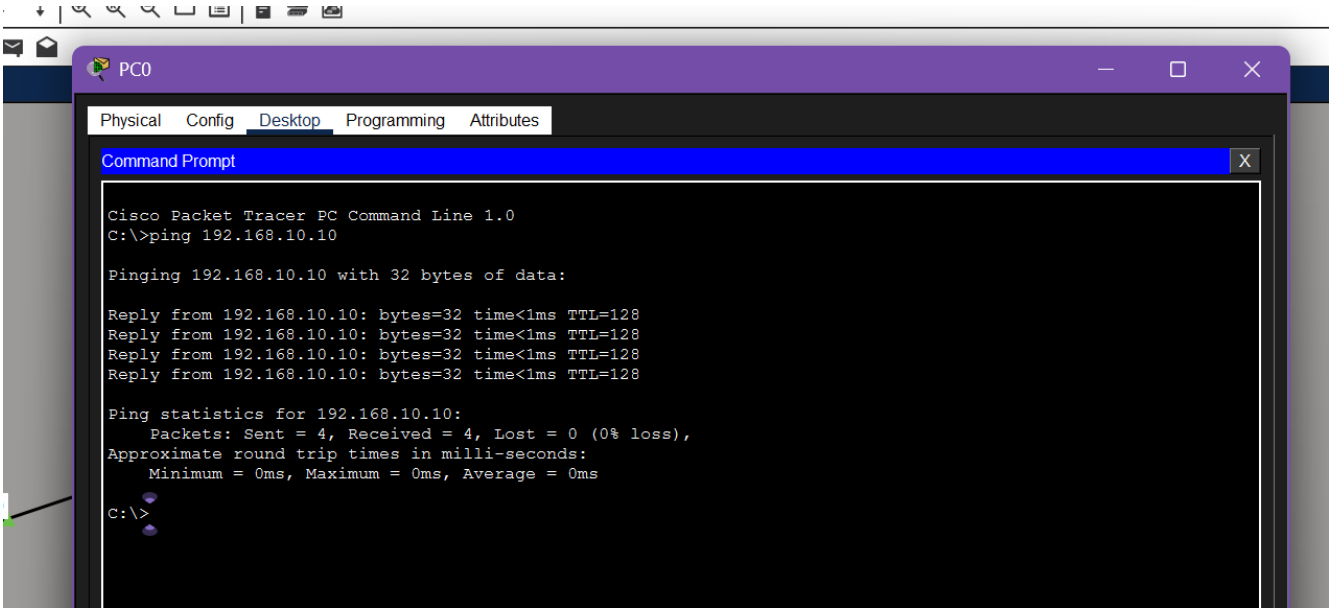
Task 3: LAN Topology Build

Answer : I would create the following simple Class C LAN in Cisco Packet Tracer with 5 systems and 1 network printer.



I would connect all six devices to a switch using copper straight-through cables. I would then configure the IP address and subnet mask on each device.

- PC1 → Desktop → IP Configuration → 192.168.10.2 / 255.255.255.0
- PC2 → Desktop → IP Configuration → 192.168.10.3 / 255.255.255.0
- PC3 → Desktop → IP Configuration → 192.168.10.4 / 255.255.255.0
- PC4 → Desktop → IP Configuration → 192.168.10.5 / 255.255.255.0
- PC5 → Desktop → IP Configuration → 192.168.10.6 / 255.255.255.0
- Printer → configure the network settings as 192.168.10.7 / 255.255.255.0



For testing, I would open the Command Prompt on each PC and use ping commands such as ping 192.168.10.3 and ping 192.168.10.7. All devices should reply successfully. If a ping fails, I would check the cable, interface status, IP address and subnet mask.

Task 4: Clean OS Installation

Answer : I would perform the clean installation using a Linux ISO in VirtualBox or on the workstation.

1. I would create a new virtual machine or boot the workstation from the installation ISO.
2. I would select the required language, keyboard layout and installation option.
3. I would choose a clean/custom installation instead of an automatic upgrade.
4. I would manually create the required disk partitions. I would check the disk size before applying the changes so I do not select the wrong drive.
5. I would continue the OS installation and allow the system to restart.
6. After the first boot, I would install required drivers and run system updates.
7. I would check the startup applications and disable unnecessary programs that do not need to start with the OS.
8. I would check unnecessary services carefully and avoid disabling services required by the operating system or security software.
9. I would restart the computer and confirm that the OS boots without errors.
10. I would check the relevant Linux system tools to confirm that startup usage has improved.

SECTION C – MINI CAPSTONE PROJECT

Mini Project: Nexus Innovations – Diagnosing Random System Reboots

Scenario : Three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets.

1. Three Hardware Components I Would Check First

If the systems pass POST but restart during heavy workloads, I would first check the following three hardware areas:

- 1. CPU/GPU cooling system – Rendering creates high CPU and GPU load. If airflow is poor, temperatures can rise until the system becomes unstable or protection mechanisms cause a shutdown/restart.
- 2. PSU – A rendering workload can increase power demand. If the PSU is overloaded, aging or unstable, the system may restart when the CPU and GPU draw more power.
- 3. RAM and motherboard power delivery – Unstable RAM or motherboard power delivery can cause crashes under sustained load. I would check RAM seating, memory errors and motherboard temperature/diagnostic information.

2. Why the Systems Pass POST in the Morning but Fail in the Afternoon

POST happens when the computer starts and checks the basic hardware. Passing POST does not mean the system will remain stable under a long heavy workload.

In the morning, the systems may be cool because they have not been under load. During rendering in the afternoon, the CPU and GPU generate much more heat. Because the workstations are inside poorly ventilated cabinets, hot air can build up and the cooling system may not remove heat effectively.

If temperatures become too high, the CPU or GPU can reduce performance through thermal throttling, and continued overheating can result in instability or a restart. A PSU problem may also appear only under higher load because power demand is much greater during rendering.

3. Stress Test to Confirm a Thermal or Power Issue

I would use a suitable CPU/GPU stress-testing tool and monitor temperatures, clock speeds, system stability and power-related symptoms.

1. Record the normal idle CPU and GPU temperatures before starting the test.
2. Run a controlled CPU stress test and monitor the temperature.
3. Run a GPU stress test separately and monitor the temperature.

4. If possible, run a combined CPU and GPU workload for a controlled period while watching temperatures and system behaviour.
5. Stop the test if temperatures reach an unsafe level or the system becomes unstable.
6. If the system restarts when temperatures rise sharply, that would support a thermal cause. If the restart happens mainly when both CPU and GPU power demand is high while temperatures remain reasonable, I would investigate the PSU and power delivery.

4. Three Specific Remediation Steps

1. Improve airflow – Move the workstations out of the enclosed cabinets or provide proper intake and exhaust airflow. I would also clean dust from fans and heatsinks.
2. Check and improve cooling – Verify that CPU/GPU fans are working, check the CPU cooler mounting and thermal paste, and confirm that case airflow is not blocked.
3. Check power and hardware stability – Test the PSU, inspect power connectors, test RAM, and monitor CPU/GPU temperatures during the workload. Replace a faulty or insufficient component if testing confirms the problem.

5. PSU Upgrade Justification

A PSU upgrade might be necessary if the existing PSU does not provide enough wattage or stable power for the CPU and GPU under full rendering load. I would first check the PSU's rated capacity, age, efficiency and required power connectors and compare this with the system's actual hardware requirements.

However, I would not replace the PSU only because the computers are restarting. If temperature testing shows that the CPU or GPU is overheating because of the poorly ventilated cabinets, improving cooling should be the first action. The PSU should be upgraded when testing or power calculations show that the current unit is insufficient or unstable.

SECTION D – AI-AUGMENTED LEARNING

STEP 1 – BUILD WITH AI

Exact Prompt Given to the AI

> *Create an IP addressing and subnetting plan for Nexus Innovations' startup office network covering 5 systems and 1 network printer. Specify the subnet, subnet mask, usable host range, broadcast address, an IP address for each of the 5 systems and the printer, and identify which addresses should be reserved for future expansion. Use a simple Class C private IPv4 network and show the calculations clearly.*

AI's Original Subnetting Plan

The initial plan used a /29 network. A /29 provides 8 total addresses, 6 usable host addresses, so it is enough for the current 5 systems and 1 printer but leaves no usable address for future expansion.

Item	AI Original Plan
Network	192.168.10.0/29
Subnet Mask	255.255.255.248
Usable Host Range	192.168.10.1 – 192.168.10.6
Broadcast	192.168.10.7
System 1	192.168.10.1
System 2	192.168.10.2
System 3	192.168.10.3
System 4	192.168.10.4
System 5	192.168.10.5
Printer	192.168.10.6
Future Expansion	No unused usable addresses

My Verification and Corrected Version

I checked the network and host calculation. The /29 network is mathematically valid for the current six devices, but it does not provide any free usable address for future expansion. Since the assessment specifically asks for addresses to reserve for future expansion, I changed the network to /28.

Item	Corrected Plan
Network	192.168.10.0/28
Subnet Mask	255.255.255.240
Usable Host Range	192.168.10.1 – 192.168.10.14
Broadcast	192.168.10.15

System 1	192.168.10.2
System 2	192.168.10.3
System 3	192.168.10.4
System 4	192.168.10.5
System 5	192.168.10.6
Printer	192.168.10.7
Reserved for Future Expansion	192.168.10.8 – 192.168.10.14
Optional Gateway Reservation	192.168.10.1

Subnetting Check

- 192.168.10.0/28 contains 16 total addresses.
- Network address = 192.168.10.0.
- Usable host addresses = 192.168.10.1 through 192.168.10.14, giving 14 usable hosts.
- Broadcast address = 192.168.10.15.
- The plan uses six device addresses and keeps additional addresses available for future expansion.

What I Changed and Why

I changed the subnet from /29 to /28 because the original /29 plan had exactly six usable host addresses, which were all used by the five systems and one printer. It was correct for the current devices but did not leave any space for future expansion. The /28 plan provides 14 usable addresses, so the required devices can be assigned addresses while several addresses remain reserved for future use.